Abdul Maajid Aga

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Education

B.Eng. (Honours) Robotics, Autonomous & Interactive Systems Engineering **Sep. 2023 – May 2026**
First Class Honours, Heriot-Watt University *Dubai, United Arab Emirates*

Work Experience

Research Assistant | *Heriot-Watt University* **September 2026 – Present**
Cog Waters Project

- Developing the embedded and mechatronic systems of a marine **Unmanned Surface Vehicle**, designing a waterproof electronics enclosure and sealing the vehicle for sea deployment.

Intern | *ABB Robotics* **August 2026 – September 2026**
Robotic Sorting System

- Built a sorting pipeline for a dual-arm **ABB YuMi** robot with an **Intel RealSense** camera, using a **Small Language Model (SLM)** to turn plain text commands (e.g., *"sort the red cubes from the yellow cylinders"*) into step-by-step instructions.
- Used **Grounded SAM2** to detect and segment objects from the camera feed, combined this with depth data to find each object's 3D position, then used **PyRANSAC3D** on the point cloud data (PCD) to identify shapes for sorting.
- Used **cuRobo** for fast, collision-free motion planning, turning the SLM's instructions into pick-and-place moves for the robot.

Robotic Chess

- Built an interactive human-vs-robot chess system, using an **ABB GoFa CRB 15000** cobot connected to the **Stockfish** chess engine.
- Wrote a **Python** backend to turn chess moves into real-world coordinates, sending them to the robot controller through a custom **RAPID** socket server, and calibrated the `wobjdata` reference frames and Z-offsets for accurate movement.
- Built an **OpenCV** vision pipeline to track piece moves with **100% accuracy**, and added an API-based state sync in **RobotStudio** to handle cases where the camera view was blocked during motion.

Military Desert Rescue Vehicle | *R&D Intern, AIMBeyond Lab* **February 2026 – April 2026**

- Researched **Drive-by-Wire (DbW)** conversion setups and compared vendor costs to pick the steering and braking hardware for a military ATV.
- Drew detailed electrical schematics in **KiCad** to plan how the chosen actuators would connect to an **MRS PLC** gateway over the **J1939** protocol.
- Built system flowcharts for the self-driving navigation stack, combining **LiDAR** and stereo camera data across the sensor fusion and perception layers for rough desert terrain.

Projects

Formula Student AI | *Autonomous & Control Systems Engineer, Atlas Racing* **May – Jul 2025 & May – Jul 2026**

- Built custom vehicle nodes in **CARLA** and **IPG Automotive**, combining **LiDAR**, stereo camera, and **INS** data over **ROS 2** to tune **FAST-SLAM** (tested against **ORB-SLAM** and **RTAB-Map**) and add **Pure Pursuit**/**PID** control for accurate, full-circuit self-driving.
- Built a **C++ CANopen** interface with under 10ms latency to connect to the Autonomous Driving System (ADS).
- Led testing and diagnostics on the driverless stack using **Nav2** and sensor calibration in simulation, getting **safety certification** for high-speed track events.

High-Voltage EV Powertrain | *Powertrain Systems Engineer* **December 2025 – January 2026**

- Built a 72V high-voltage powertrain in a one-month sprint, making a custom energy storage system from reconditioned **NiMH** cells to give a stable DC bus for a 3-phase AC motor.
- As part of a 3-person team, wired a **Curtis motor controller** into the vehicle harness to handle ignition, forward/reverse drive states, and live dashboard telemetry, keeping the whole system in sync and safe.

Generative Modelling for Stroke Rehabilitation | *Honours Dissertation* **September 2025 – July 2026**

- Compared generative models (**GAN**, **Transformer**, **LSTM-VAE**) in **PyTorch** on the **MHH dataset** to work around limited clinical data, and used a tuned **CVAE**.
- Fed the generated, severity-graded motion data into a detailed **MyoSuite (MuJoCo)** simulation of the human upper limb's muscles and bones.
- Checked the full pipeline's biomechanical accuracy using **Inverse Kinematics (IK)**, **Inverse Dynamics (ID)**, muscle synergy extraction, and simulated **EMG** co-contraction analysis.

Waste Collection Robot | *Project Lead* **September 2024 – February 2025**

- Designed a custom **4-DOF** robotic arm and wrote **Inverse Kinematics (IK)** and path planning code in **ROS 2** for accurate movement.
- Led a 6-person team through the full development cycle, adding **Cartographer**-based Visual SLAM through **RGB-D** cameras and a **Raspberry Pi 5** for self-driving navigation and finding waste targets.

Marine Unmanned Surface Vehicle | *Embedded Systems Lead* **July 2024 – July 2025**

- As part of a 9-person team in the **Mubadala** competition, built a **micro-ROS** setup on **ESP32** microcontrollers, first over **SPI**, then switched to **UART**, and wrote custom **C++** libraries to handle real-time I/O and expose thrusters, pumps, and sensors as **ROS 2** nodes to an **NVIDIA Jetson**.
- Added safety watchdogs for **PWM**-controlled actuation in case of communication loss, and made the power setup more stable by isolating high-current motors to cut signal interference in marine environments.

Skills

Software & Programming: C++, Python, ROS/ROS 2, Nav2, micro-ROS, PyTorch, MATLAB/Simulink, RAPID, Linux, Git
Simulation & Validation: CARLA, IPG Automotive, Gazebo, MuJoCo, MyoSuite, Webots, RobotStudio, Sim2Real & HIL Workflows
Perception & Autonomy: SLAM (FAST, ORB SLAM, RTAB Map), Sensor Fusion, Localization, Path & Motion Planning (CuRoBo, MoveIt), Computer Vision, Inverse Kinematics/Dynamics
Embedded & Hardware Systems: NVIDIA Jetson, Raspberry Pi, ESP32, Custom PCB Design (KiCad), Real-Time I/O, Motor Control, High-Voltage Systems, Sensor Integration
Controls & Communication: CAN/CANopen/J1939, PLC, Drive-by-Wire, SPI/UART, PID Control, Pure Pursuit, Safety-Critical Systems

Publications

Simulating an Autonomous System in CARLA using ROS 2 | *IEEE (ICSPIS 2025)* **November 2025**

- Developed a modular autonomous racing stack utilizing ROS 2 to fuse 3D LiDAR and Stereo Camera data for high-speed trajectory optimization and obstacle avoidance within the **CARLA** environment.

SUSHI: A Vision System for Reactive ASV Navigation | *Journal of Intelligent & Robotic Systems* **March 2026**

- Engineered a vision-based navigation framework for Autonomous Surface Vehicles (ASVs) using Multi-Field Path Planning to enable reactive, uninformed exploration in complex marine environments.

Achievements

Formula Student AI Static Winner **2025**
Institution of Mechanical Engineers (IMechE) *Silverstone, UK*

- Secured **1st place** in Overall AI Statics and the Real World AI Award (perfect 100/100 score) against 25+ teams, becoming the first GCC team to win two major awards.

IET Student Prize Winner **2024**
Institution of Engineering & Technology *International*

- Awarded for outstanding academic achievement, innovation, and exceptional potential within the global engineering community.

University Prize **2024**
Heriot-Watt University *Dubai, UAE*

- Awarded to the **highest-ranking School of Engineering student** across the global campuses (Dubai, Malaysia, UK) for exceptional academic performance.